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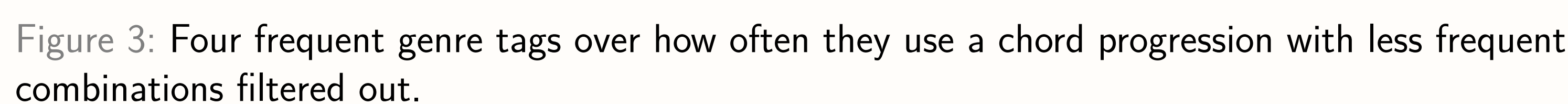
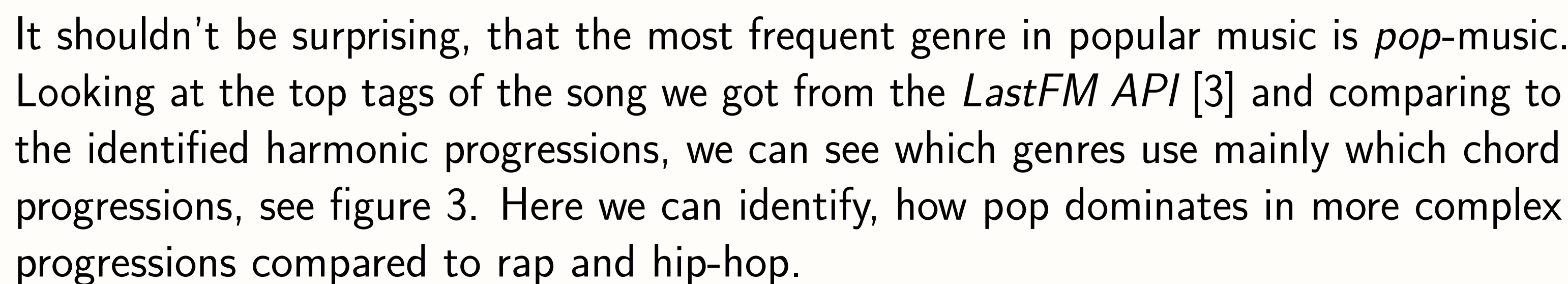
– Billy Joel

Not only is it interesting to look at popular music from the point of view of musicians aspiring to create the next hit single. Indeed, it is also interesting to analyse recent music since music is an essential part of any culture. By analysing popular music one can learn about what the current culture is, how it changed and how it varied. In the following, we'll look at the most popular music of the last two decades [1] to answer which musical and lyrical elements we can identify and analyse the change of them.

What main chord repetition should you use to write the perfect song? By webscraping *Ultimate Guitar* [5], we found the most frequently used harmonic progressions of the last 20 years, see figure 1.

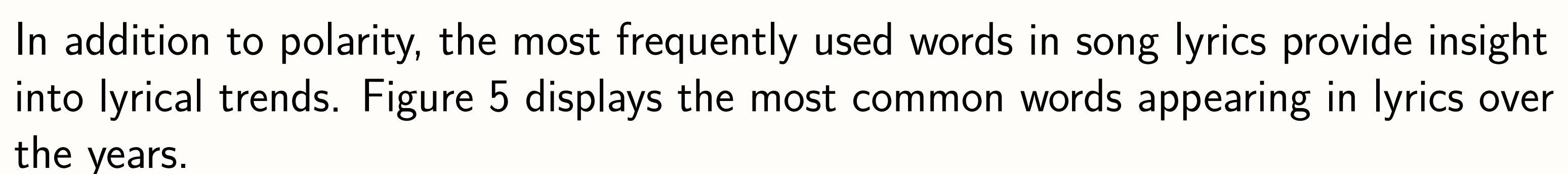
Harmonic Progression	Absolute Frequency
(1, 6, 3, 7)	52
(1, 5, 6, 4)	39
(1, 4)	36
(1, 1, 1, 1)	30
(1, 5, 3, 2)	19
(4, 1)	16
(1, 5)	16
(1, 6, 4, 5)	14
(1, 7)	14
(1, 3, 6, 4)	14

Musicians often write songs in the same key they can sing in and therefore use the same chords more often. Most chords in figure 2 can be played in C-major or A-minor. We can also see that A-minor key is more popular than C-major key, since D and Em work better in A-minor compared to C-major.¹



¹D and Em are roughly the IV and V of A-minor, but the II and III of C-major,

Song lyrics often reflect emotions and moods of their time. By extracting them from *Genius API* [2] and analyzing their polarity over the years with *TextBlob* [4], we can identify cultural trends about how the language in songs had become positively or negatively charged, see figure 4.



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